Nine national providers
ARSO · CHMI · FOEN · GKD Bayern ·
LUBW · NVE · Rijkswaterstaat · SYKE · USGS
River-only, complete-year filter
lakes, seas, canals, estuaries, deltas out
a year with any missing day: no truth
fewer than four complete years: out
146 networks · 457 stations ·
5,114 completely observed station-years
Split at the network level
never at the station-year level;
stratified on provider × donor regime
110 development · 36 reserved
Reserved panel: retrospective holdout
28 of 36 networks have earlier
programme use; not an unused
confirmation set for new analyses
Outages designed, configuration frozen
51,524 cases; empirical and stratified arms
40,704 development · 10,820 reserved
declared before the reserved panel opened
matched retention; at least five points;
a network-clustered interval excluding zero
Reserved panel opened once
9.7% → 2.5% at 70% retention
–7.13 points [–8.97, –5.36]
the empirical arm is unresolved
Sealed USGS confirmation
39 HUC4 networks · 91 stations ·
8,973 designed cases, outside the
development programme; scored once
–7.9 points [–10.2, –5.8]
Prior-access audit and clean subset
24 of 39 networks appear in earlier work
15 outcome-clean networks: –9.4 points
[–12.4, –6.0]; a transparency diagnostic,
not new confirmation
Pre-registered transport panel
SMHI · UK EA · Hub'Eau: 32 primary
networks across three contexts,
four assurance strata combinations,
sealed and scored once: –4.6 points
[–6.5, –3.3]
Primary and secondary strata
quality-checked stratum: –6.2 points
[–8.4, –4.5], agreeing with the pooled
verdict; NVE secondary stratum,
21 networks: –6.1 points [–8.7, –4.6]
Truth-free deployment
2,422 real incomplete station-years,
which carry no truth; the frozen
screen is applied to them unchanged
Two admission rules compared
length rule vs peak-window screening
disagree on 15.1% of the 2,422 years;
the certificate instead supports abstention
where the interval straddles the threshold